

Removing Global Gluing: A Multiverse Base with a Global Countable Core, Probability and Processes without a Global Context

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Abstract

Kolmogorov’s probability, ZFC and the Church–Turing thesis share a tacit principle: compatible local data glue uniquely into a global object in a single global context. We drop this principle and make its failure measurable. The base theory B describes a family of ZFC worlds under forcing and rank extensions, without amalgamation, with one strong axiom: the countable core is invariant under extension. We determine its strength, classify global and local mathematics, prove a stabilization theorem that classifies ZFC-independent statements, and compute the ordinal cost of amalgamating worlds. On this base, probability takes laws as global and samples as local, and processes are studied through the circumstances of questioning. For partially exchangeable processes the hidden global parameters are exactly asymptotic rates, present iff synchronization is scale-free, and we prove rigidity in the complementary regime for broad classes of synchronizations.

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1 Introduction

1.1 The wall and the principle behind it

In machine learning, theoretical computer science and the foundations of physics, one meets the same obstruction. Once a description stops presupposing a single global context—one universe of sets, one sample space, one time line, one scalar objective—it seems to become *indescribable*.

- Reward-free, memory-organizing continual learners have no scalar metric, and reintroducing one reintroduces the abandoned objective.
- Relational physics must decide what a theory predicts, evolution or structure.
- Experiment and theory are locked in a loop: triggers are designed from predictions, predictions from triggered data.

We call the common tacit principle GLU_0 : *there is a single global context, and in it compatible local data glue uniquely into a global object*. It has four clauses:

1. uniqueness;
2. determinate existence;
3. well-founded shape;
4. non-unique existence, i.e. choice.

Its instances include Kolmogorov’s extension theorem [41], extensionality plus power set in ZFC [36], hidden-variable models [5], and “the state at the next step” of a mechanism [21]. GLU_0 can live on three layers (n58):

Layer	In ZFC / Kolmogorov	In the base $\mathbf{B}$ below
Objects (completed totalities, choice objects, sample spaces)	global	local
Descriptions of nature (families of laws on contexts)	forced to glue	glue only when they glue
Mathematical possibility (do all worlds amalgamate?)	one universe	possibilities converge (S4.2), objects need not

The walls in the motivating fields sit on the first two layers.

1.2 Goals

The goals fixed at the outset were:

- (R1) make “a global model exists” a proposition that can be false, and measure how false it is;
- (R2) use a measurement-centred language in which the set-theoretic apparatus sits at the top;
- (R3) recover ZFC and standard measure theory where appropriate;
- (R4) explain, and where possible decide, statements that ZFC cannot decide;
- (R5) keep the axioms interpretable;
- (R6) replace linear time, absolute causation, reductionism and scalar objectives by statements with precise ranges of validity.

1.3 Results in brief

- **Base** (§2). $\mathbf{B}$ is a set theory of worlds and extensions without amalgamation. Its axiom M4 says literally that *records are not disturbed by later recording*.
 - The bottom layer consists of countable records, which are global; the top layer of completed totalities and choice objects, which are local; the two are connected by recording, i.e. extension.
 - Strength: three tiers $\mathbf{B}^- \subset \mathbf{B} \subset \mathbf{B}^+$.
 - Global and local mathematics are classified; facts glue while existence does not; globality comes in three strict levels.
 - The failure of amalgamation is measured by an ordinal, computed exactly.

- **Stabilization** (§3). A question with finitely many changes of answer along a recording system is a finite Boolean combination of persistent answers. Under confluence it has a unique limit.
 - This extends Ershov’s hierarchy from linear to branching time.
 - It classifies ZFC-independent statements into four kinds.
- **Probability** (§4). Laws are global and samples local; events are descriptions.
 - A family of laws on contexts has a global law iff it can be realized consistently; the obstruction is finite and arithmetic.
 - The fate of old point sets under extension is determined.
- **Processes** (§5).
 - Records are exactly the public and undisturbed behaviours.
 - The cost of hiding a failure equals the number of residual behaviours.
 - Facts see structure, not position; linear time is an index of individuation.
 - De Finetti and Bell are two ends of one theorem.
 - The hidden global parameters of partially exchangeable processes are exactly asymptotic rates, present iff synchronization is scale-free.
- **Rigidity** (§6). There are no other hidden parameters for broad classes of synchronizations:
 - concave envelopes, including the critical case;
 - arbitrary windows of bounded ratio;
 - all i.i.d. random windows;
 - spikes and plateaus;
 - three chains with any synchronization graph.
- **A conservative language layer** (§7). The gluing scheme $\text{GLU}(\pi)$ adds kinds of objects with non-classical value, shape and context. It is conservative over ZFC. Choice, excluded middle and Kolmogorov’s theorem become theorems whose validity depends on the parameter.
- **Readings** (§8). The results translate into precise statements about triggers in physics, scalar metrics in machine learning, and computation as stabilization.
- **Contributions** (§9). What is new against prior work, why it matters, and what it offers the applied sciences in expression, tools and understanding.

A unifying shape. Separately proved results share one form. The *global* content of a situation is what is invariant under all individuations:

- under all worlds, it is the countable core;
- under all positions, it is the structure;
- under all linearizations, it is the trace together with the rates.

The classical demands—one universe, one evolution, one sample space, one scalar—are each a demand to make a choice.

Conventions. Each claim carries a status tag:

- [known]: a known result placed in our setting;
- [new]: a result of this work;
- [new org.]: known ingredients organized in a new way.

References of the form (n97) point to the working notes described in §12.

2 The base theory

2.1 Language and axioms

There are two sorts, *sets* and *worlds*, and three primitive relations:

- $x \in y$;
- $x \varepsilon w$: the set x is in the world w ;
- $u \preceq v$: the world v extends u .

We write $w \models \varphi$ for φ relativized to w .

M1 Every world is transitive and satisfies ZFC; every set lies in some world. By transitivity all worlds share ω and the hereditarily finite sets (ω -*standardness*).

M2 $\preceq$ is generated by two kinds of step:

- *width*: $u \preceq u[G]$ for a generic G ;
- *height*: $u \preceq v$ when $u = V_\alpha^v$.

M2a': intermediate models of width steps are worlds.

M3 Every world has a P -generic extension for each poset P in it, and a proper height extension.

M4 For $u \preceq v$ and parameters $a \in H(\omega_1)^u$: $H(\omega_1)^u \models \varphi(a) \iff H(\omega_1)^v \models \varphi(a)$.

M4⁺ The same with $L(\mathbb{R})$ and real parameters.

No amalgamation (common extension of two worlds) and no world containing all worlds are assumed. This is the precise sense in which GLU_0 is dropped. Worlds are not sets, so there is no outer universe inside the theory. We set $\mathbf{B}^- = \text{M1–M3}$, $\mathbf{B} = \text{M1–M4}$ and $\mathbf{B}^+ = \mathbf{B} + \text{M4}^+$.

Remark 2.1 (Reading of M4). Forcing adds a record, a reading of a source that the world could not answer. M4 is the principle “statements about existing records are not disturbed by later recording”, applied to the countable core (n53). Recording is thus a primitive relation of the base, and the measurement-centred language of goal (R2) is realized as extension between worlds rather than as a probabilistic replacement of $\in$.

2.2 Strength

Theorem 2.2 (Strength).

1. M_4 holds iff every world satisfies two-step projective generic absoluteness PGA_2 . [new, routine]
2. B is equiconsistent with ZFC plus infinitely many strong cardinals [39]. [known]
 - Lower bound: Hauser [30].
 - Upper bound: Woodin showed that n strong cardinals give two-step Σ_{n+3}^1 absoluteness after a collapse (see [62]); compactness then finishes.
3. M_4 implies that every set has a sharp, that every Σ_2^1 set is universally Baire [17, 61], and hence $V \neq L$ in every world. M_4 does not imply PD: the global layer contains statements that are global yet independent. [known ingredients]
4. B^+ proves $\text{AD}^{L(\mathbb{R})}$ in every world. A proper class of Woodin cardinals gives a model; at least an iterable inner model with ω Woodin cardinals is required [57, 44]. [known]
5. B^- requires only enough transitive models, roughly a proper class of worldly cardinals (below one inaccessible).

All large-cardinal strength comes from M_4 .

Remark 2.3. The results relevant to the original goal need only B^- or a single world. These are:

- the global/local classification, the modal logic, coexistence and amalgamation;
- all probability and process results.

M_4 is needed exactly where “everything definable from reals is regular” is used (Theorem 2.6, and the two-valued fate of definable records).

2.3 Global and local mathematics

A statement is *global* if its truth is the same in all worlds that contain its parameters.

Theorem 2.4 ([known facts [36, 4], new org.]; (n55), (n57)).

1. Size. Countability is persistent, and every set becomes countable in some extension. Hence countability (and finiteness) are the only global size facts.
2. Totalities. For infinite x , some extension adds a subset of x . So $\mathcal{P}(x)$, $\mathbb{R}$ and 2^ω are local. This is GLU_0 at infinity.
3. Choice. For a given family, a choice function persists. But choice objects lose their defining property in extensions:
 - a Cohen real splits every old infinite set, so an old ultrafilter is no longer an ultrafilter;
 - an old Vitali set misses the class of a new real;
 - an old well-order of $\mathbb{R}$ omits the new reals.

Choice holds in each world, but what it chooses is not global.

4. σ -algebras. As families of point sets they are local; as algebras of codes they are global.
5. Laws. Borel codes and their measures are absolute [56, 55], so laws are global. Random points over w are not in w , so samples exist only in extensions. Kolmogorov's framework is the special case "fix one world".
6. CH can be forced both ways over every world. It is local: it asks about a completed totality.

Theorem 2.5 (Limit and records; (n59), (n44)). *Along a chain of worlds in which every set is eventually collapsed:*

- the countable cores form an elementary chain;
- their union is a model of ZFC^- with collection in which every set is countable (the theory T_0 of an earlier formulation);
- in this limit, choice holds and every set of reals is null.

Moreover, M_4 implies the "staging principle" (C): for every real a , $L[a] \cap \mathbb{R}$ is countable.

Theorem 2.6 (Regularity; (n95)). *In $\text{B}+M_2a'$, in every world every projective set with real parameters is Lebesgue measurable and has the Baire property. [new, standard argument]*

- Sketch.*
1. Take the Boolean value of $\varphi(\dot{r}, a)$ in the random algebra of w ; it is a Borel code b .
 2. Collapse the continuum of w . In the collapse, every real random over w generates an intermediate world, which by M_4 agrees with the collapse.
 3. So in the collapse the symmetric difference of the set and b is contained in the non-random reals, hence null.
 4. This is a projective statement, so M_4 brings it back to w .

□

2.4 Modal structure and the two borderlines

Theorem 2.7 ([known] [27, 26, 25]; (n56), (n57), (n71)). *The modal logic of extension, for sentences with parameters, is S4.2: objects diverge, possibilities converge. The notes also argue that this persists when width and height steps are combined (this combination is not in the literature).*

Two borderlines organize the global structure (n71), (n72):

- *Well-foundedness governs possibility.* Non-well-founded top extensions give S4 [28, 26]. Along such extensions the countable core is unchanged, so the modal logic on the core is trivial and all forks act above the countable core (n76).
- *ω -standardness governs the finite layer.* Records, frequencies and contextuality are the same in all worlds. The price is a height obstruction between worlds.

The height obstruction is not removed by non-well-founded ω -models.

Theorem 2.8 ((n71)). *If N is an ω -model of ZFC containing reals x, y , then the well-founded part of N has height $> \omega_1^{x \oplus y}$.*

In the non-well-founded variant D (which allows top extensions): with M_4 , $0^\#$ freezes the theory of L , and Σ_1 assertions converge in a collapse. Without M_4 , the principle .2 fails from worlds satisfying $V = L$. With M_4 , .2 holds on the frozen and Σ_1 fragments (n101), (n104).

2.5 Coexistence and levels of globality

Theorem 2.9 ([new]; (n81), (n74), (n86), (n89)).

1. Every coexistence pattern occurs. *Let $c_1, \dots, c_n$ be mutually generic Cohen reals over w and K a simplicial complex on $\{1, \dots, n\}$ containing all singletons. There is a model of $\mathbf{B}$ in which $\{c_i : i \in F\}$ lie in a common world iff $F \in K$.*
2. The only constraint is closure under definability: *if b is definable from a by a code in the core, every world containing a contains b .*
3. Facts glue, existence does not. Δ_0 facts about finitely many sets are the same in all worlds containing them. The global $\in$ satisfies extensionality and foundation, but $\mathbf{B}$ has no global pairing. This is dual to contextuality: coexistence without sharing versus sharing without coexistence.
4. Amalgamation is independent of $\mathbf{B}$ (non-amalgamable Cohen extensions go back to Mostowski; see [24]). There are sparse models with sets that never coexist, and directed models in which all worlds amalgamate. Adding amalgamation restores GLU_0 at the level of objects.
5. Three strict levels: $\mathbf{B} \subsetneq \mathbf{B} + \text{global pairing} \subsetneq \mathbf{B} + \text{amalgamation}$.
 - Under amalgamation, the global sets form a T_0 universe.
 - The classical single universe contradicts $M3$ and returns in no strengthening.

Theorem 2.10 (Absorption; (n95), (n96)). *Suppose a world w has a height extension N with a Woodin cardinal above $o(w)$, and N is iterable in the outer universe. Then any two generic extensions of w have a common extension, at a greater height. [known technique: generic iteration]*

So in the natural interpretation, world-level non-gluing is a same-height phenomenon, and $\mathbf{B}$ is realism about countable sets.

2.6 The amalgamation height

For worlds w_1, w_2 , let $\text{amh}(w_1, w_2)$ be the least height of a transitive ZFC model containing both.

Theorem 2.11 ([new]; (n119)). *If $w = L_\alpha[A]$ with $A \subseteq \alpha$ and $A \in w$, then*

$$\text{amh}(w[c], w[d]) = \min\{\beta \geq \alpha : L_\beta[A, c, d] \models \text{ZFC}\}.$$

Proof. If $M \supseteq w[c] \cup w[d]$ has height β , then $L^M[A, c, d] = L_\beta[A, c, d] \models \text{ZFC}$ and $\beta \geq \alpha$. Conversely, $L_\beta[A, c, d]$ contains w and c, d , hence every valuation of names from w . $\square$

Corollary 2.12.

1. *Same-height amalgamation holds iff $L_\alpha[A, c, d] \models \text{ZFC}$. Mutually generic pairs amalgamate at the same height; Mostowski-type pairs [24] do not.*
2. *If $\omega_1^{c \oplus d} \geq \alpha$ and $A = \emptyset$, then amh is the least ZFC-height of $L[c \oplus d]$, independent of α .*
3. *Over a countable L_α , the heights for pairs of Cohen reals are unbounded in ω_1 (take $d = c \oplus_{\text{XOR}} z$).*

For general worlds (n124), amh lies between $\sup_{A \in w} \min\{\beta \geq \alpha : L_\beta[A, c \oplus d] \models \text{ZFC}\}$ and $\min\{\beta : L_\beta(w \cup \{c, d\}) \models \text{ZFC}\}$. The earlier lower bound $\sup \omega_1^{x \oplus y}$ (n71) is far weaker. The gap in the bracket disappears whenever w is generated by a set parameter; ground-model definability [45, 19] is the relevant tool for generic extensions of such worlds. World-level non-gluing is therefore measured by an ordinal, determined by the join of the information.

3 Stabilization along recording

Definition 3.1.

- A *recording system* is a preorder $(W, \leq)$, and a *question* is a map $P : W \rightarrow \{0, 1\}$.
- $\text{Alt}(P, k, w)$ holds if some chain of extensions starting at w changes the answer k times. The *change number* is $c(P, w) = \sup\{k : \text{Alt}(P, k, w)\}$.
- A set S is *persistent* if it is closed upward. A persistent answer is a *button*.
- The axiom .2 for P is $\Diamond \Box (P = b) \rightarrow \Box \Diamond (P = b)$.

Theorem 3.2 (Stabilization, [new]; Lean `Stabilize`; (n122), (n123)).

1. If P is determined by n buttons, the number of possible changes is at most the number of unpushed buttons.
2. (No confluence needed.) Let $V_j = \{w : \neg \text{Alt}(P, j, w)\}$ and let Y_k be the upward closure of the stages where exactly k changes remain and $P = 1$.
 - These sets are persistent.
 - At a stage with exactly k remaining changes, $P = 1$ iff the stage lies in Y_k .

So $c \leq N$ makes P a Boolean combination of $2N + 2$ persistent sets.
3. (Confluence.) Under .2 the limit value is unique. If moreover $c(P, w_0) \leq N$, then above w_0 the answer is the limit value flipped once for each remaining change: a combination of N buttons and a constant.

On linear stages, (3) is Ershov’s theorem that n -c.e. sets are Boolean combinations of c.e. sets [15, 16], with limits given by Shoenfield’s limit lemma [54]; the same notion appears as limiting recursion and identification in the limit [22, 53, 23]. The theorem shows that linear time is inessential and confluence is what matters. Without confluence the “limit” is the persistent set Y_0 , which records the path taken. Learning along non-exchangeable question trees is of this kind (§5).

Set theory. Under extension, .2 holds [27].

- M4 puts every core sentence at level 0 (Lean `WorldsStable.core_level_zero`).
- Every finite level occurs. Take the ratchet $b_i = “\aleph_i^w \text{ is countable}”$ ($i \leq n$) and let φ_n say that an odd number of the b_i hold; then $c(\varphi_n) = n$.
- Sentences forced both ways from every world, such as CH, are switches with $c = \infty$ (Lean `WorldsStable.switch_all_levels`).

- Height steps can raise levels. “There is an inaccessible” is a negative button under width steps and a switch once height steps are added.

Proposition 3.3 (Four kinds of independence, [new]). *A ZFC-independent sentence is, in $\mathcal{B}$, one of the following.*

1. Decidable: $\mathcal{B}$ decides it (e.g. “every set has a sharp”).
2. An oracle bit: $\mathcal{B} \vdash \Box\varphi \vee \Box\neg\varphi$ without deciding φ (e.g. PD, which $\mathcal{B}^+$ decides).
3. Limit-decidable: finite change number, with a unique limit value supplied by recording.
4. A switch: no answer (e.g. CH).

Their computational analogues are:

1. computable;
2. computable in a fixed unknown oracle;
3. a finite Ershov level;
4. non-stabilizing.

Strengthening axioms reads off oracle bits: this is the consistency-strength ladder.

4 The probability layer

4.1 Laws, descriptions and samples

Recall the basic notions:

- a *law* is a probability measure given by countable data;
- an *event* is a *description*, i.e. a Borel code (projective under M4);
- a *fresh sample* over w is a point, in an extension, that is random over w .

Theorem 4.1 ([new org.]; (n61), (n43), (n45); Lean Askable, Sources).

1. Measure belongs to descriptions.
 - Measures of descriptions are invariant under extension.
 - Every local point set becomes null in some extension.
 - Non-measurability is local and temporary.
 - In the limit T_0 , choice coexists with “every set of reals is null”.
2. Fresh samples avoid every local point set almost surely; only descriptions can be hit.
3. Independence is extension: a second sample is fresh over the world containing the first.
4. Askable questions. A question can be answered to arbitrary precision by finitely many readings of a source iff it is measurable. In particular, every question about a record is askable.

5. Records of sources. (*Independent rereading plays the role of Larsen–Skou testing [43].*) A source has a record iff two independent readings agree on their first n answers with probability bounded below uniformly in n . Hence records are the atoms, and atomless sources (continuous randomness) have laws but no records.
6. “A global model exists” has two independent meanings: a global law exists, or a global law has a record. Each has its own measure of failure. Relations can be records while their ends are not: synchronous coupling makes “equal” certain without either side being a record.

4.2 Contextual laws

Theorem 4.2 ([new org.], cf. [1, 5]; (n61), (n63), (n68); Lean Records, Exchange). *Consider a family of laws on overlapping contexts, compatible on overlaps.*

1. *It has a global law iff it has a realization agreeing on all overlaps with probability one. The maximal probability of agreement equals the noncontextual fraction.*
2. *The obstruction is finite. If no global law exists, there is $\eta > 0$ such that every shared record table of any length deviates by at least η on some context.*
3. *The existence of a global law is arithmetic, hence the same in all worlds.*
4. *Contextuality and amalgamation of worlds are independent in both directions.*

There are thus two failures of gluing: one finite (values: contextuality) and one infinite (worlds: amalgamation height).

4.3 The fate of old point sets

Theorem 4.3 ([new org./new]; (n67), (n70), (n75), (n79), (n84); Lean Fate, Kill). *Let $w \preceq w'$.*

1. Duality.
 - *If w' contains a Cohen real over w , then $\mathbb{R}^w$ is null in w' .*
 - *If w' contains a random real over w , then $\mathbb{R}^w$ is meagre in w' .*
 - *Random extensions preserve outer measure exactly; Cohen extensions preserve non-meagreness.*
 - *$\mathbb{R}^w$ is either null or of full outer measure in w' .*
2. Characterization. *A record of ideal type I kills A iff A is I -a.e. covered by a Borel null family. Measure-carried ideals preserve outer measure exactly.*
3. Killing without Cohen reals. *An infinite set unsplit by the reals of w (for instance a Mathias real [46]) kills all of $\mathbb{R}^w$. Every Mathias–Prikry forcing kills everything.*
4. Stratified sets. *For $A = \bigcup_i (\mathbb{R}^{v_i} \cap B_i)$ one has $\mu^*(A)^{w'} = \mu(\bigcup\{B_i : v_i \text{ not killed}\})$. So intermediate values arise from which world a point comes from.*
5. Dichotomy. *For good records (Π_1^1 -on- Σ_1^1 ideal forcings and their finite iterations), each old set’s measure is either fully kept or entirely lost. In $\mathbf{B}$ this holds for every record definable from reals; partial loss requires parameters that are local choice objects.*

6. Window. *Either $\mathbb{R}^w$ is null in w' , or $\mu_*(A)^w \leq \mu^*(A)^{w'} \leq \mu^*(A)^w$. The uncertainty of measure is exactly non-measurability.*

Theorem 4.4 (Realized measure extensions; (n100), (n105)).

- *The effect of an extension on old point sets is a translation-invariant extension of Lebesgue measure, given by traces of new Borel sets.*
- *It takes only the values 0 and 1 on invariant sets.*
- *It is always separable, so non-separable invariant extensions [38] are never realized.*
- *It grows monotonically along branches.*
- *Covering codes typical for a Borel measure cover only old null sets.*

A strict interior value of the window arises iff a Szpilrajn-type extension [59] is realized, for instance by selectively killing the reals of a smaller world.

Theorem 4.5 ((n107)). *A real that guesses all ground finite sequences on intervals (for example, a Cohen real) kills $\mathbb{R}^v$ together with the reals of every random extension of v . Hence “Cohen, then random” does not realize an interior value; compare [50].*

Question 4.6 (Q**; (n109)–(n118)). *Let m be Mathias over v and r random over $v[m]$. Is $\mathbb{R}^{v[r]}$ non-null in $v[m][r]$?*

- Equivalently (using the decomposition of Mathias forcing [46] and Galvin–Prikry [20]): with U the generic Ramsey ultrafilter, $V' = v[U]$ and F the filter generated by U in $V'[r]$, does Mathias–Prikry forcing $\mathbb{M}_F$ preserve positive outer measure of the ground reals?
- Known constraints:
 - no guessing reals are added;
 - covers bounded by ground functions fail;
 - ground-coded measures and measures built from bounded tuples of m cannot witness a positive answer via the Kakutani class [37];
 - the literature on Mathias–Prikry forcing [34] characterizes dominating and eventually different reals, not measure; F is a non-meagre P-filter in Talagrand’s sense [60].
- Conjecture: for non-meagre P-filters, killing happens only through the “eventually constant along the generic” mechanism; hence Q** holds.

5 The process layer

5.1 Behaviours and the four circumstances

A *behaviour* maps question histories (who asked what, in what order) to answers or answer laws. A classical *record* is an answer invariant in four circumstances:

- asking again;
- who asks;

- what was asked before;
- what is asked together.

The classical foundations assume that every answer is a record.

Theorem 5.1 ([new org.]; (n47)–(n49); Lean Circumstances, Individuation).

1. *A behaviour has a record model iff it is public and undisturbed.*
2. *Hidden-state models always exist, and the minimal number of hidden states is the number of residual behaviours (a Myhill–Nerode form [48, 49]). The degree of failure is what a classical description must invent.*
3. *The four failures are pairwise independent.*
4. *Facts see structure, not position. Two behaviours have the same invariant facts iff they have the same residual behaviours.*
 - *For undisturbed behaviours, facts determine everything.*
 - *Two flip-memories have the same facts yet differ.*
 - *Position is a fact iff no other residual behaviour can do the same things. In particular, it is a fact when disturbance is irreversible.*
5. *Prediction is the task that fixes each answer: structure plus position. Relational tasks are not predictions.*
6. *Individuation. Writing the circumstance into the question gives every behaviour a record model; this is needed only when the behaviour is not public or is disturbed.*
 - *Linear time is the index that individuates repeated asking.*
 - *“Who asks” has the same form. Two objects’ facts through viewpoints coincide iff their sets of viewpoints coincide, and which viewpoint one occupies is not a fact.*

Theorem 5.2 (Evaluation; (n53); Lean Circumstances). *An evaluation that can be redone at any time gives a behaviour the same value as anything it can become. For an irreversibly changing system it therefore sees only the terminal structure; seeing present ability requires individuation by time, which cannot be redone. For recurrent systems there is no such trade-off.*

Theorem 5.3 (Networks; (n51); Lean Network). *With at least two askers, a public and non-interfering behaviour is undisturbed, hence has a record model. In a public world, disturbance appears as interference.*

5.2 Order, time and learning

Theorem 5.4 ((n62), (n63); Lean Protocols).

1. *The future of a behaviour is independent of question order iff questions pairwise commute. Undisturbed behaviours commute. Recorded answers and future behaviour give two independent kinds of order dependence.*
2. *A mixture of records read freshly is order-independent (the Pólya urn is the example; cf. de Finetti [8, 32]). Hence order-dependent behaviours are not hidden records read freshly.*

3. *Along a non-commuting question tree, each path is a classical martingale; unasked branches are never revealed; the learning limit depends on the strategy. Only in the exchangeable case is the limit strategy-independent.*

5.3 Partial exchangeability, de Finetti and Bell

Let the alphabet be partitioned into classes that commute with each other.

Theorem 5.5 ([new]; (n65); Lean Exchange).

1. *Two words have the same trace iff they have the same stream in every class. A trace is a tuple of streams; a word is a schedule together with the streams.*
2. *A law is trace-exchangeable iff it is a random tuple of streams interleaved by an exchangeable schedule. Given the streams, the schedule is uniform over permutations with the given counts.*
3. *Who decides which stream to read.*
 - *The process itself: arbitrary streams interleaved by an exchangeable schedule; de Finetti and arbitrary processes are the two ends.*
 - *The position: Diaconis–Freedman [10].*
 - *The askers: the trace law is exchangeable iff classes do not signal. A hidden global structure exists iff GLU holds; it always exists when at most one class has a choice, and may fail with two (PR boxes [52]).*

De Finetti and Bell are the two ends of one theorem.

Theorem 5.6 (Order spectrum; (n66); Lean Spectrum). *Let a behaviour be class-exchangeable.*

1. *The future depends only on within-class order, and every tuple of within-class orders is realized. The order spectrum is the image of the group of within-class permutations, and this upper bound is attained.*
2. *Every model reaches distinct states for distinct spectrum members: $\text{states} \geq \text{residuals} \geq |\text{Spec}|$.*
3. *Under uniformly random order, the within-class orders are independent and uniform, so order entropy adds over classes.*
4. *Order sensitivity satisfies a triangle inequality and is Lipschitz in within-class Kendall distance.*
5. *Coarse-graining preserves exchangeability and only shrinks the spectrum. An order-independent metric cannot detect whether the evaluated system is order-independent.*

5.4 General independence and hidden rates

For a general independence relation, traces are Mazurkiewicz traces [47, 12] and the dependence graph has components.

Theorem 5.7 ((n69), (n73); Lean Traces).

1. *Trace-exchangeability reduces to the components of the dependence graph: components are interleaved by an i.i.d. schedule given rates, and each component stream is invariant under swapping adjacent independent letters.*

2. *Infinite traces almost surely have no maximal element.*
3. *Every central exhaustive linearization is a mixture of: a rate vector π on components; after finite time, i.i.d. interleaving at rates π ; and fresh local shuffles.*
4. *Within a component, every surviving rate vector of the chains yields an extreme central linearization, and distinct rates yield distinct ones. Under geometric synchronization a whole interval of rates survives, so finite incomparability does not imply rigidity.*
5. *Infinitely many complete cuts, or uniformly bounded incomparability, imply rigidity.*

Consider two chains synchronized at checkpoints: at the k -th checkpoint the position difference must lie in a window of width W_k , and P_k is the elapsed time. Put $\rho_k = W_k/P_k$.

Theorem 5.8 (Hidden rates, [new]; (n83), (n88)).

1. *If $\liminf \rho > 0$, the surviving rates form a non-degenerate interval: there is a continuous hidden rate.*
2. *If $\liminf \rho = 0$, only the rate $\frac{1}{2}$ survives.*
3. *Learning.*
 - *Counts (the trace) are sufficient.*
 - *The posterior is consistent.*
 - *$\sqrt{n}(i_n/n - p) \Rightarrow N(0, p(1-p))$, exactly as without synchronization.*

Synchronization decides whether rates exist, not how precisely they are learned.

4. *Several chains. Connected and dense synchronization gives unique rates. Scale-invariant edges give an open rate polytope, whose dimension is the number of hidden parameters (n108).*

6 Rigidity

Rigidity means that no hidden parameters exist beyond the trace, i.e. the central linearization is unique. It is the complementary question to Theorem 5.8.

6.1 Reduction to a killed walk

Theorem 6.1 ((n78), (n93), (n94); Lean Sandwich).

1. *The one-step checkpoint kernel is explicit (Chu–Vandermonde): $K_k(t, t') = \binom{f(k+1)+f(k)+1-t+t'}{f(k+1)}$.*
2. *One-dimensional reduction. After a gauge change the checkpoint process is the checkpoint chain*

$$D_{k+1} = D_k + W_{k+1} - \text{NB}(W_{k+1}, \frac{1}{2}),$$

killed outside $[1, W_{k+1}]$. In “X-time” this is a zero-mean walk with variance $2W$ per checkpoint, and the gap equals the window.

3. *Its kernels are totally positive (TP₂) [40]. Hence the Birkhoff projective diameter [6] of a block equals the log cross-ratio of the four corners.*

4. Four equivalent forms. *Rigidity* $\iff \Delta(M_{K \rightarrow L}) \rightarrow 0$ for every $K \iff$ the corner cross-ratio tends to 1 $\iff$ the survival-conditioned laws from the bottom and top corners merge in ratio distance.
5. It suffices to find disjoint blocks with $\sum_j e^{-\Delta_j/2} = \infty$ (Birkhoff contraction).
6. Plateau theorem. Constant windows W_j on plateaus of length at least $m(W_j)$ give rigidity, even with unbounded incomparability; numerically $m(W) = O(W)$.

6.2 The phase transition

Theorem 6.2 ([new]; (n97), (n98), (n108)). *If the windows have a concave envelope (up to $O(\sqrt{W})$), then rigidity holds iff $\rho \rightarrow 0$. This includes the critical case $\omega \asymp P/\log P$ and oscillating W^2/P . Combined with Theorem 5.8: the hidden parameters are exactly the rates, and rates exist exactly when synchronization has no characteristic scale.*

Proof architecture.

- A three-block sandwich reduces the diameter to entrance, middle and exit estimates (Lean Sandwich).
- *Slowly varying windows* ($W^2 = O(P)$) use:
 - an embedding into X-time;
 - a ruin bound for monitoring only at checkpoints (dip-and-return);
 - pinned local lower bounds and tubes along lines (a uniform local limit theorem for non-identical steps [51]);
 - a piecewise exponential tilt changing only at checkpoints, whose cost is controlled by the total variation of slopes.
- *Wide windows* use the corner formula, the Holley inequality [33] for rows, an exact constant tilt, and a boundary layer at the top corner.
- The general concave case chooses good non-doubling scales via the tangent intercept $b(P) = \omega - P\omega'$.

6.3 Irregular windows

Theorem 6.3 ([new]; (n109), (n117), (n123), (n124)).

1. Bottlenecks. *The one-step diameter has an explicit formula; $\sum_k e^{-\Delta(q_k)/2} = \infty$ implies rigidity for any windows. For example, double bottlenecks, or logarithmically small windows, occurring infinitely often suffice.*
2. Bounded ratio. *If infinitely many runs of $\lceil \Theta W \rceil$ checkpoints have $W \leq W_j \leq \Gamma W$ (Γ fixed, W the minimum on the run), the heap is rigid, whatever the shape of the windows.*
3. Random windows. *All windows $\omega_j U_j$ with U_j i.i.d. and bounded are almost surely rigid.*
4. Spikes and plateaus. *Let the windows between two levels $\leq \Gamma W$ rise to a plateau of X-time $T \geq \eta^2 W^2$ with all windows $\geq C\sqrt{T}$. Then the kernel across the plateau has bounded diameter. Arbitrarily high spikes therefore act as resets and help rigidity.*

Key steps of (2). Envelope. Take the lower Lipschitz envelope $\hat{\omega}_t = \min_j (W_j + C_0|t - j|)$ with $C_0 = 1 + \Gamma/\eta^2$. Runs of checkpoints more than ηW above $\hat{\omega}$ are shorter than $2\eta^2 W$ (**Lean Envelope.run_bound**). After piecewise-linear smoothing, $\hat{\omega}$ is a tilted envelope with bounded tilt cost and with contacts every $O(\eta^2 W)$ checkpoints.

Upper bound. Survival from depth δ is at most $C(\delta + \ell)/W$ with $\ell = 2\eta W$. This follows from counting how often a coarse chain visits the region above 2ℓ (arcsine law); each visit costs a constant killing probability.

Entrance ratio. Stop at the first time σ the walk is ℓ deep.

- Harris–FKG [29, 18] bounds the survivors that stay shallow: survival is decreasing and “stay above” is increasing.
- A last-passage decomposition bounds the expected depth.

The short-time survival factor then appears in numerator and denominator and *cancels*. The floor is never reached before σ . $\square$

Proof of (4). The kernel is squeezed between $\frac{1}{2}q_T$ and q_T , where q_T is the half-line kernel, which has product form $\asymp (x \vee \sqrt{g})(y \vee \sqrt{g})/T^{3/2}$. The factor $\frac{1}{2}$ holds because the bridge rarely exceeds $C\sqrt{T}$ (**Lean Closure.plateau_cross**).

Recursively, a proof of rigidity can fail only for “fractal” window profiles, stuck at every scale between the reset and the bounded-ratio regimes.

6.4 Several chains

The state is the vector of differences along a spanning tree of the synchronization graph, and the window is a polytope.

Theorem 6.4 ((n109), (n110), (n112), (n116), (n123)).

1. Exact decomposition (*three chains on a path*): $D^1 = U - D^2/2$, where U and the $(2, 3)$ -shuffle are independent. Strong scale separation ($W^2 \lesssim \sqrt{W^1}$) reduces to the one-dimensional theorems, recursively for paths and trees.
2. Plateau and contraction criteria are insensitive to dimension.
3. Three chains at comparable scales, any synchronization graph (*path or triangle*) are rigid.
 - After whitening, the unit steps are three vectors at 120° , and the window is a parallelogram or hexagon with 60° and 120° corners.
 - At 60° corners $h = xy(x + y)$ is an exact martingale of the sampled chain for every gap (**Lean Corner**).
 - At 120° corners $r^{3/2} \sin(3\theta/2)$ is harmonic up to relative error $O(1/|x|)$, by induction over dyadic annuli; the third cumulant of a sum of c steps grows like c . (This follows the harmonic-function method of Denisov–Wachtel [9], whose asymptotic results are not needed.)
4. Weyl chamber [known] [42]. For m chains the all-low corner is $\{x_1 > \dots > x_m\}$, and the Vandermonde product is exactly harmonic (**Lean** for $m = 4$). Mixed corners for $m \geq 4$ are polyhedral cones with dihedral angles $60^\circ, 90^\circ, 120^\circ$; only the 120° edges are singular.

Numerics.

- Block diameters were computed for regular, random, multiscale, spiked and oscillating windows and for three- and four-chain synchronizations (triangle included); all are bounded and decreasing.
- Entrance ratios lie in $[0.29, 0.49]$ uniformly for $K \leq 400$.
- The martingale defect at 60° corners is exactly 0; at 120° corners, the relative error times $|x|$ is ≈ 0.14 for both gaps.

7 The gluing scheme: a conservative language layer

Before arriving at **B**, the project built a scheme in which GLU_0 is relaxed parameter by parameter. It is conservative over **ZFC** and remains a language for objects outside the classical wall.

Parameters. A piece of local data has three aspects, $\pi = (W, S, \mathcal{C})$:

- its *value* W : a sharp yes/no, or a weight, possibly with names;
- its *shape* S : well-founded, or arbitrary;
- its *context* $\mathcal{C}$: a small category whose arrows are refinements and whose automorphisms are symmetries.

The classical origin is $\pi_0 = (\text{sharp}, \text{well-founded}, \text{point})$.

The scheme. $\text{GLU}(\pi)$ states that the kind U_π is the terminal object among small π -systems:

- every small π -system has exactly one solution;
- every object is the image of some node.

First-order form: images are solutions; every object is an image; images agree iff a π -bisimulation connects the nodes. For shapes this is Aczel’s anti-foundation [2]. At the origin, $\text{GLU}(\pi_0)$ is clauses 1–3 of GLU_0 . Clause 4 (choice) is asserted only at the origin.

Theorem 7.1 ([new org.]; (n24)–(n41); Lean Shapes, Kinds, Values, Instances, Over, Symmetric, Contexts, Weights, WeightedGlue, WeightedKind, RandomChoice).

1. Construction and conservativity.

- *Every U_π is built from records as pointed small systems modulo π -bisimulation, so it is unique up to isomorphism.*
- *Adding $\text{GLU}(\pi)$ proves no new statement about records.*
- *Constructed in Lean: anti-founded shapes; context kinds over arbitrary small categories with names; value kinds over every complete value structure (using only `Quot.sound`); values over preordered contexts.*

2. Identity is π -bisimulation. It reduces to extensionality on well-founded shapes, and to having a common image (Larsen–Skou bisimulation [43]) for weights. Two shapes cannot share one membership relation, so extension proceeds by adding kinds.

3. Logic. *Excluded middle holds exactly where a context cannot be refined further; double negation holds everywhere. With extensional identification, choice forces excluded middle (Diaconescu [11]).*
4. Choice criterion.
 - *Local selections glue when there is a root, without using choice.*
 - *On disjoint contexts they glue exactly when classical choice holds.*
 - *Common refinements and symmetries can block gluing.*
 - *On preorders, local existence always glues iff every connected component has a least context.*
5. Existence is local. *There are global objects with members in every context but no global member (refinement and symmetry forms). In particular, Russell's pair of socks is a global object with no global sock.*
6. Random choice without the axiom of choice. *"A random sock" is a global object whose weight is forced by symmetry to be $\frac{1}{2}$ (and $\frac{1}{n}$ for n symmetric nodes).*
7. Weights. *Existence is the shadow of weight, and shadows forget multiplicity. On directed contexts the weighted layer is Kolmogorov's theorem: global weights are unique and independent local weights glue to products. Refinement forces the value algebra $\mathbb{R}_{\geq 0}$, and regularity of a measure is exactly the survival of additivity.*
8. Origin inside every kind.
 - *Along W and C there is an embedding and a shadow, whose composite is the identity: extension is refinement.*
 - *Along S there is no shadow: cyclic objects cannot be flattened into well-founded ones, so linear time cannot be recovered by forgetting.*

Theorem 7.2 (Computation and processes as objects; (n39), (n40); Lean Local, Observe, Views, Processes).

1. *Gandy's principles for mechanisms [21] are GLU at the origin.*
2. *Deterministic local algorithms see only objects, and cannot elect a leader in a symmetric system [3, 63].*
3. *What they see depends on the value structure: sets versus counts [31].*
4. *Coins break symmetry [35].*
5. *On finite systems, two nodes are the same object iff every deterministic local algorithm keeps them in the same state (views stabilize to Larsen–Skou bisimulation).*
6. Processes. *Without observation all Markov processes are one object. With stopping as observation, stopping probabilities are properties of the object and are preserved by merging. Objects are finer than stopping laws: they remember when randomness was decided. Cyclic symmetry forces the uniform law.*

8 Readings for physics, machine learning and computation

These are precise statements placed in the motivating problems. They are not physical theories, benchmarks or architectures.

8.1 Physics

- *Structure versus evolution.* An evolution is structure plus an individuation of position. Position is a fact only where disturbance is irreversible.
- *Hidden variables.* A hidden global structure exists iff local laws glue. De Finetti and Bell are the two ends of this one theorem, and the failure η is finite and world-independent.
- *Emergent rates.* Clocks synchronized only at events have relative rates as their only global parameters beyond causal order. These rates are determined iff $\rho \rightarrow 0$.
- *Interference.* In a public world, disturbance appears as interference.

Proposition 8.1 (Triggers; [known: Solovay tests [13]]; Lean `Closure.trigger_blind`). *Let triggers T_n be coded in w with $\sum \mu(T_n) < \infty$. A phenomenon random over w lies in only finitely many T_n .*

Rare-event triggers designed within a theory therefore confirm only what the theory can describe. A phenomenon fresh relative to the theory is caught at positive frequency only by theory-independent sampling.

8.2 Machine learning

- Redoable evaluations see only terminal structure, so continual learning faces a genuine trade-off.
- Order spectra and order entropy are non-scalar invariants.
- The cost of hiding order dependence is at least the size of the order spectrum.
- Autoregression is the choice of a linear individuation. For partially exchangeable streams, the global parameters to learn are the dependence graph, the per-component laws, and rates only under scale-free synchronization.

Proposition 8.2 ([known [58, 14, 7]]; Lean `Closure.le_iff_all_linear_ext`). *Order learners by inclusion of the facts they maintain.*

1. *A faithful scalar metric exists only if this order is total.*
2. *The order is the intersection of its linear extensions.*
3. *The least number of scalars representing it is its order dimension.*

“Give one number” asks for a choice of linear extension, and “one number cannot measure intelligence” notes that this choice is not determined by the facts. The content common to all admissible rankings is the partial order. The earliest tools of the project were consistent with this view: the contextual fraction, Hodge decomposition of rankings, and the Blackwell order.

8.3 Computation

1. *Stabilization along recording* (§3). Turing computability, limit computability [22, 53] and generic absoluteness are one notion at three levels, and it requires confluence, not linear time. In this reading, ZFC-independence is non-convergence of answers along recording.
2. *Minimal hiding*. A machine hides history-dependence in state, and its minimal size is the number of residuals. Zielonka’s theorem [64] glues global trace behaviour from local components. The rate theorems identify the only non-local asymptotic parameters.
3. *Gluing parameters*. Values, symmetry and context determine which tasks local mechanisms can solve.

9 Contributions and significance

Much of the material rests on a large body of prior work. We therefore state separately what is contributed and what is only re-used.

9.1 What is new, measured against prior work

Foundations. *Prior work* includes:

- the set-theoretic and generic multiverse [25, 57];
- the modal logic of forcing and potentialism [27, 26];
- generic absoluteness and its strength [17, 30, 44, 61];
- non-amalgamable extensions [24];
- set-theoretic geology [19].

Our contribution:

1. A two-sorted axiomatization $\mathbf{B}$ derived from one principle—records are not disturbed by later recording. We give its strength in three tiers and use the global/local classification as the organizing theorem of a foundation. The mathematical strength of $\mathbf{B}$ coincides with known generic absoluteness; the novelty here is the role these facts play, not the facts.
2. The coexistence theorem (every simplicial complex is realized) and the strict three-level hierarchy of globality.
3. The exact amalgamation height, a new measure of world-level non-gluing.
4. Stabilization theorems for arbitrary recording systems. These extend Ershov’s hierarchy [15] from linear time to confluent branching time, and without confluence give a $2N + 2$ representation.
5. The resulting four-way classification of ZFC-independent statements: decidable, oracle bit, limit-decidable, switch.

Probability. *Prior work* includes:

- forcing and measure [56, 4];
- Mathias-type forcing [46, 34];
- invariant extensions of Lebesgue measure [59, 38];
- contextuality [1].

Our contribution:

1. Laws global and samples local, as theorems of the base.
2. Askability equals measurability, and records of sources are characterized by rereading.
3. The contextual obstruction is finite and arithmetic, and independent of the amalgamation of worlds.
4. A characterization of the fate of old point sets: duality, the ideal-covering criterion, the stratified formula, two-valuedness for definable records, and the window.
5. Separability and monotonicity of realized measure extensions.
6. A reduction of the remaining question Q^{**} to Mathias–Prikry forcing over a random extension.

Processes. *Prior work* includes:

- exchangeability [8, 32, 10];
- traces [47, 12];
- minimal automata [48, 49];
- Bell-type non-locality [5, 52].

Our contribution is a single framework—the four circumstances of asking—in which the following are proved:

1. records are exactly the public and undisturbed behaviours;
2. the cost of hiding a failure is the number of residual behaviours;
3. facts determine structure but not position, and linear time is an index of individuation;
4. de Finetti and Bell are the two ends of one gluing theorem;
5. the order spectrum is quantitatively described;
6. trace-exchangeable processes are represented for general independence relations;
7. hidden rates undergo a phase transition, and the learning theory of rates is given.

Random walks killed at checkpoints. These are, to our knowledge, the most substantial new theorems.

1. The rigidity phase transition (Theorem 6.2).
2. Rigidity for arbitrary windows of bounded ratio, through two devices introduced here: the lower Lipschitz envelope, and the cancellation of short-time survival in entrance ratios.
3. Rigidity for all i.i.d. random windows.
4. Spikes and plateaus act as resets.
5. Rigidity for three chains with any synchronization graph.

The tools—Birkhoff contraction [6], total positivity [40], Holley and FKG inequalities [33, 18], and cone harmonic functions [9, 42]—are classical. Their combination and the results are new.

The gluing scheme. *Prior work* includes anti-foundation [2], probabilistic bisimulation [43], sheaf-theoretic contextuality [1], and anonymous distributed computing [3, 63, 31].

Our contribution is one parametrized, conservative scheme covering value, shape and context, with:

- a proved criterion for when local choices glue on preorders;
- embeddings and shadows of the classical origin;
- Gandy’s principles [21] identified as the origin of the scheme;
- machine-checked constructions.

Formalization. About 636 machine-checked theorems in Lean 4 cover the finite parts, with an axiom profile that tracks where choice is used.

9.2 Significance

- *For foundations.*
 - “A global model exists” becomes a graded, measurable proposition. Its failure is finite for values (contextuality), counted by hidden states for processes, a window for measure, and an ordinal for worlds.
 - Independence is explained rather than merely exhibited. CH is undecidable because it is local (a switch). PD is a global oracle bit that only stronger axioms read off. Some independent statements have canonical limit values supplied by recording itself.
- *For probability.* Kolmogorov’s framework is recovered as the special case of a fixed world. Sample spaces and non-measurable sets are local, and events are descriptions. This removes the tension between choice and measure: in the limit of $\mathbf{B}$, choice holds and every set of reals is null.
- *For process theory.* The question “what is hidden behind a partially commuting process?” has a precise answer: the trace, and rates exactly when synchronization has no characteristic scale.
- *For computation.* Stabilization, not linear time, is the notion behind decidability-in-the-limit. Confluence is exactly what is needed for answers to have a unique limit.

9.3 Contributions to the applied sciences

The applied value lies in precise problem statements, computable quantities and explanations, not in new physical theories, benchmarks or architectures.

Expression. The work supplies a vocabulary with exact definitions for describing systems without a global context:

- the four circumstances of asking;
- records versus sources;
- structure versus position;
- traces and rates in place of a single time line;
- global versus local quantities;
- the partial order of maintained facts in place of a scalar score.

Tools. Each of the following quantities can be computed for a concrete system:

1. the *cost of hiding*: the minimal number of hidden states needed for a classical model;
2. the *noncontextual fraction* η of a family of observations;
3. the *order spectrum and order entropy* of a learner or process;
4. the *order dimension* of a task family, which is the number of metrics needed to compare systems faithfully;
5. the *stabilization level* of a question under a recording protocol, which says whether the answer converges and after how many changes;
6. the *Birkhoff diameter* of block kernels, a numerical test of whether hidden rates in a multi-stream system are identifiable;
7. a *trigger criterion*: summable, theory-designed triggers catch only finitely many theory-fresh events.

Understanding.

- *Machine learning.* The “single scalar” loop is a demand for a choice of linear extension. Redoable evaluation of an irreversibly learning system can only see its terminal structure. Autoregression is one linear individuation of a partially exchangeable stream, whose genuinely global parameters are the dependence graph and, under scale-free synchronization, rates.
- *Physics.* What a relational theory can predict are invariants under individuation (structure). Evolution adds a position, which is a fact only for irreversible disturbance. Rare-event triggers designed by a theory are blind to what is fresh relative to it. Relative rates of event-synchronized clocks are emergent global parameters, present iff synchronization is scale-free.
- *Computation.* Local mechanisms compute what the gluing parameters allow. Beyond linear time, decidability in the limit needs confluence.

Whether these quantities are useful in practice must be established by practitioners on concrete systems; the present work provides their definitions and theorems.

10 Open problems

1. Q^{**} : Mathias, then random.
2. Rigidity for fractal window profiles; this is a form of the monotonicity conjecture that shrinking windows creates no hidden parameters.
3. The uniform cone estimate near 120° edges, which would give rigidity for $m \geq 4$ chains at comparable scales.
4. Whether the lower bound of the amalgamation bracket is attained for general worlds.
5. The modal logic of extension with M4 beyond the frozen and Σ_1 fragments. There is a conditional counterexample via the core model K in a non-well-founded variant.
6. Exact strength of B^- ; a reference for S4.2 when width and height steps are combined.
7. A probabilistic stabilization theorem for learning systems.

11 Status of the claims

Claim	Status
Global/local; S4.2; limit T_0	complete (standard forcing)
Strength of B , B^+ ; $M4 \Leftrightarrow PGA_2$	complete (cited)
Strength of B^-	approximate
Projective measurability; absorption	complete (cited techniques)
Coexistence, pairing, amalgamation levels	complete
Amalgamation height	complete; general case bracketed
Stabilization theorems; stabilization in B	complete, Lean
Probability layer	complete (cited forcing facts; partly Lean)
Measure fate, window, realized extensions	complete
Q^{**}	open
Process layer (circumstances, facts, networks, order, traces)	complete, largely Lean
Hidden rates; learning	complete
Rigidity: reduction, criteria, phase transition	complete (cited: local limit theorem, Karlin–Rinott, Holley)
Rigidity: bottlenecks, bounded ratio, random, spikes/plateaus	complete on top of the one-dimensional lemmas
Three chains, comparable scales	complete (constants at 120° not all written)
Scale separation for paths and trees	sketch
$m \geq 4$ chains, comparable scales	structure; one lemma missing
Gluing scheme $GLU(\pi)$; computation and processes as objects	complete, Lean
Readings	known mathematics; readings are ours

12 Methodology

This work was developed in about seventy iterative rounds, documented in 124 working notes. The notes are cited as (nn) and are available on request. Each round began by checking that the step was driven by the guiding principle—dropping GLU_0 and measuring its failure—rather than by translating existing theories, and ended with an explicit grading of what was proved. Finite parts are machine-checked in Lean 4 (Appendix A). All rigidity statements were checked numerically on exact kernels, and load-bearing literature was verified at the end. Parts of the early drafts were written by the author independently. After many rounds of revision it is not clear how much of that text survives verbatim. Most of the later work was carried out in dialogue with the AI assistants Claude Opus 5 and Claude Opus 5.5 (Anthropic), which drafted proofs, Lean code and numerical experiments under the author’s direction. Responsibility for all claims rests with the author.

A Formalization

Two Lean 4 packages accompany the notes.

- `lean/` is mathlib-free and has 463 theorems:
 - 240 use no axioms;
 - 208 use only `propext/Quot.sound`;
 - 15 use choice, exactly where choice is the content.
- `lean-mathlib/` has 173 theorems.

Module	Content
Shapes, Kinds, Values, Instances, Over, Symmetric, Scheme, Contexts, Supports	the gluing scheme: shapes and anti-foundation, context kinds, value kinds, identity principles, choice criterion, logic by contexts
Grades, Weights, WeightedGlue, WeightedKind, RandomChoice, Socks	value algebra forced by refinement; weights as answers; Kolmogorov localized; random choice without AC
Local, Observe, Views, Processes	local computation sees objects; views; processes as objects
Records, Askable, Sources, Circumstances, Individuation, Network	records and GLU; askable $\Leftrightarrow$ measurable; records of sources behaviours, hiding cost, facts and structure, individuation, networks, evaluation
Protocols, Exchange, Spectrum, Traces	order and commutation; trace representation; PR boxes; order spectra; general independence
Modal, Worlds, WorldsStable, Stabilize	$.2 \Leftrightarrow$ confluence; shape of B and changing worlds; stabilization theorems
Fate, Kill, Window	Fubini steps of measure fate; unsplit sets kill; covering obstruction

Module	Content
Sandwich, Corner, Closure	Envelope, three-block sandwich and corner formula; Lipschitz envelopes; exact corner martingales; plateau kernels; linear extensions; triggers

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